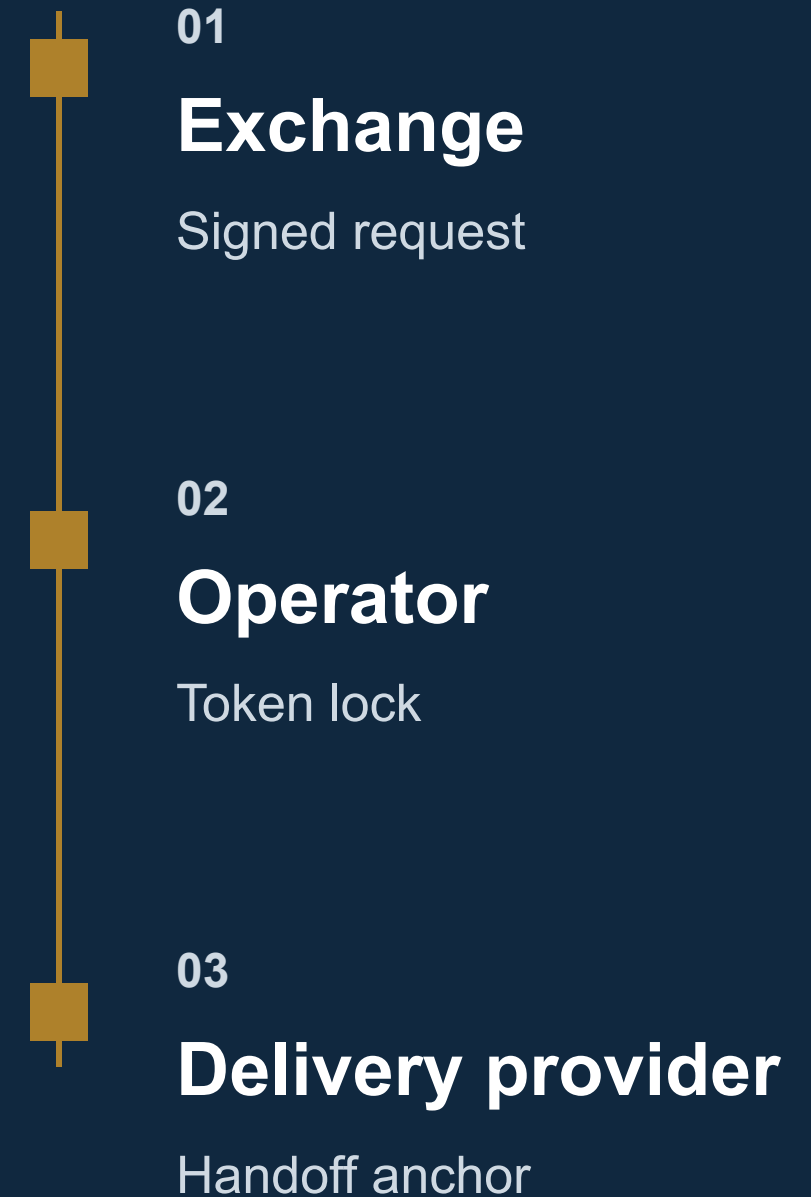


# Make every redemption delay attributable.

Independent evidence for the journey  
from a token lock to a delivery claim.

A commitment to record and prove each step.  
Redemption approval remains an institutional decision.

ESCROW / EXCHANGE INBOX / PUBLIC VERIFIER



# One redemption journey. Three accountable clocks.

01

## Locate the delay

Separate exchange forwarding, operator decision-making, and delivery-provider recording into distinct obligations.

02

## Keep evidence portable

Let holders check signed records and public commitments without relying on the institution's server.

03

## Test before adoption

Use reproducible local scenarios and verified testnet deployments as the starting point for a partner pilot.

**Implemented today: signed receipts + public log + redemption escrow + exchange inbox.**

# A token lock reveals custody— not who is delaying redemption.

Holders see one delay. Responsibility spans several institutions.



VISIBLE ONCHAIN

Lock, amounts, token movements

EVIDENCE GAP

Decision timing, handoff, delivery claims

# Start with holders who can sign and institutions that can integrate.

## PRIMARY USER

# Self-custody holders and institutions

Direct locking starts the operator's clock.  
The holder retains control of challenges,  
reclaims, disputes, and acknowledgement.

## ADOPTER

### Issuer / operator

Integration incentive: distinguish responsibility  
across the redemption chain. Hypothesis, not traction.

## CONDITIONAL

### Exchange-wallet customer

Needs an external signing key. The inbox does not  
verify customer status, balance, or eligibility.

## OUTSIDE DIRECT SCOPE

### Custodial user without a key

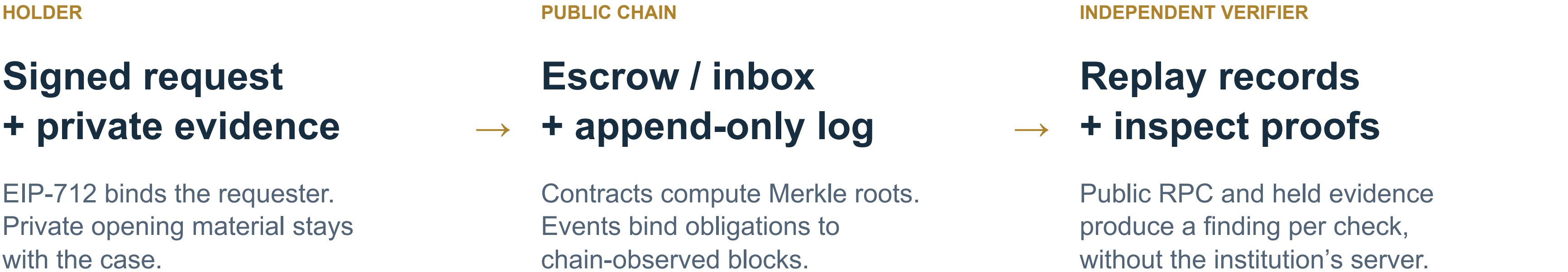
Cannot independently sign the current flow.  
Passkeys and gas sponsorship are not implemented.

# Each clock starts from a different onchain event.

| RESPONSIBLE PARTY    | CLOCK STARTS AT                          | REQUIRED EVIDENCE                      | IF UNANSWERED                                     |
|----------------------|------------------------------------------|----------------------------------------|---------------------------------------------------|
| 01 Exchange          | Holder-signed request submitted to inbox | Forward into escrow or prove rejection | Exchange breach only; operator clock not started  |
| 02 Operator          | Token lock (direct or via inbox)         | Decision record; then a proven handoff | finalize returns tokens after the response window |
| 03 Delivery provider | Handoff anchor recorded onchain          | Delivery claim in provider's own log   | STALLED; tokens remain locked                     |

Requests and locks start clocks without waiting for a separate acceptance receipt.

# Public commitments make records independently checkable.



**PRIVATE:** decision outcome + reason opening    **PUBLIC:** addresses, amounts, deadlines, states, burn / return

# Return, hold, or burn— each follows explicit conditions.

## RETURN

**Operator non-response is finalized,  
or the handoff deadline passes.**

Tokens return through finalize / reclaim.

## HOLD

**Provider non-response → STALLED.  
Holder dispute → DISPUTED.**

Tokens stay locked pending offchain  
resolution.

## BURN

**Provider delivery claim is proven;  
holder acknowledges or dispute window expires.**

After expiry in DELIVERED, anyone can  
call burn.

**The dispute window begins when the escrow accepts the proof—not at an older delivery anchor.**

# Missing evidence must remain distinct from a proven breach.

**CONFIRMED**

Signature / binding / proof checks out

**NOT\_DUE**

Deadline has not arrived

**OBLIGATION\_UNMET**

Inspect the failed check and evidence

**UNVERIFIABLE**

Evidence is insufficient

**OUT\_OF\_SCOPE**

Fact is outside the tool’s guarantees

**THREATS ADDRESSED**

Institution: omission and lateness  
Requester: fabricated obligations  
Third party: forged bundles

**OUTSIDE THE GUARANTEE**

Physical gold and actual delivery  
Truth of private reasons  
Legal enforcement and chain finality

# The prototype is reproducible; production validation is still ahead.

191

## Automated tests

107 Solidity + 84 TypeScript

30

## Local scenarios

20 attack + 10 honest

3

## Source-verified contracts

Sourcify exact\_match on Sepolia

### LOCAL EVIDENCE

All 30 scenarios matched expected findings.  
Zero expectation mismatches in honest cases.

This is not a statistical false-positive rate.

### PUBLIC EVIDENCE

Six deployment / registration transactions.  
MockGold + escrow + inbox on Sepolia.

No physical redemption or full public scenario run.

# The contribution is accountability for offchain decisions.

| REFERENCE         | CORE MECHANISM                                       | BOUNDARY / ADAPTATION                                                                            |
|-------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Rollup inboxes    | Forced inclusion                                     | The rollup protocol can enforce inclusion.<br>An offchain service cannot be forced the same way. |
| Transparency logs | Append-only commitments and inclusion proofs         | Users monitor their own entries; public roots provide evidence of inclusion.                     |
| BlockNotice       | Bound receipts, chain-derived deadlines, asset locks | Locks start the operator clock without a receipt; handoff starts the provider clock.             |

**Adapt established primitives to make delay attribution independently verifiable.**

# Adoption depends on identity, disputes, and operational controls.

01

## Identity & access

Authenticate institutions and customers.  
Address inbox spam and false balance claims.  
Plan key rotation and wallet support.

02

## Disputes & incentives

Define offchain resolution for locked cases.  
Validate issuer and legal-team incentives.  
Evaluate institutional collateral / penalties.

03

## Operational readiness

Set finality and monitoring policies.  
Audit contracts and token assumptions.  
Pilot with a real operating partner.

**Shipped: working prototype and testnet evidence. Next: validate the operating model with a partner.**

# Verify the evidence. Then test the operating model.

REPRODUCE THE LOCAL DEMO

**npm run demo**

CHECK THE PUBLIC DEPLOYMENT

**npm run check:escrow -- sepolia**

Setup and SEPOLIA\_RPC\_URL are documented in the README.

SEPOLIA / 11155111

|          |                   |
|----------|-------------------|
| MockGold | 0xb3a791...955b1a |
| Escrow   | 0x8c8cbf...f7778c |
| Inbox    | 0x870238...bb5d   |

Three services, one demo wallet.  
MockGold has no physical-gold backing.  
Source verification is not a security audit.

**[github.com/tnwjd023-boop/BlockNotice\\_GoldRWA](https://github.com/tnwjd023-boop/BlockNotice_GoldRWA)**